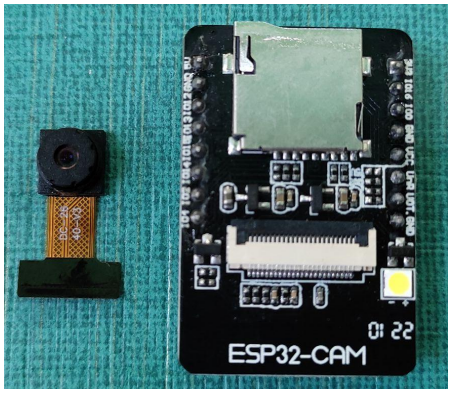
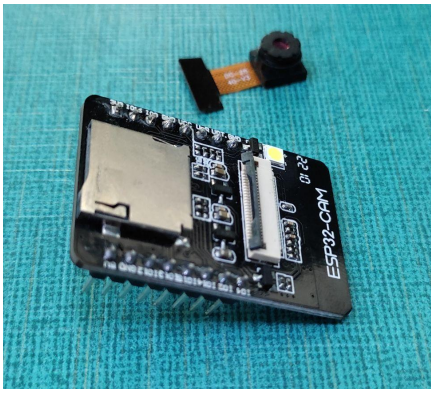
HARDWARE SETUP

To install the hardware and make it ready for firmware upload, following steps are required:

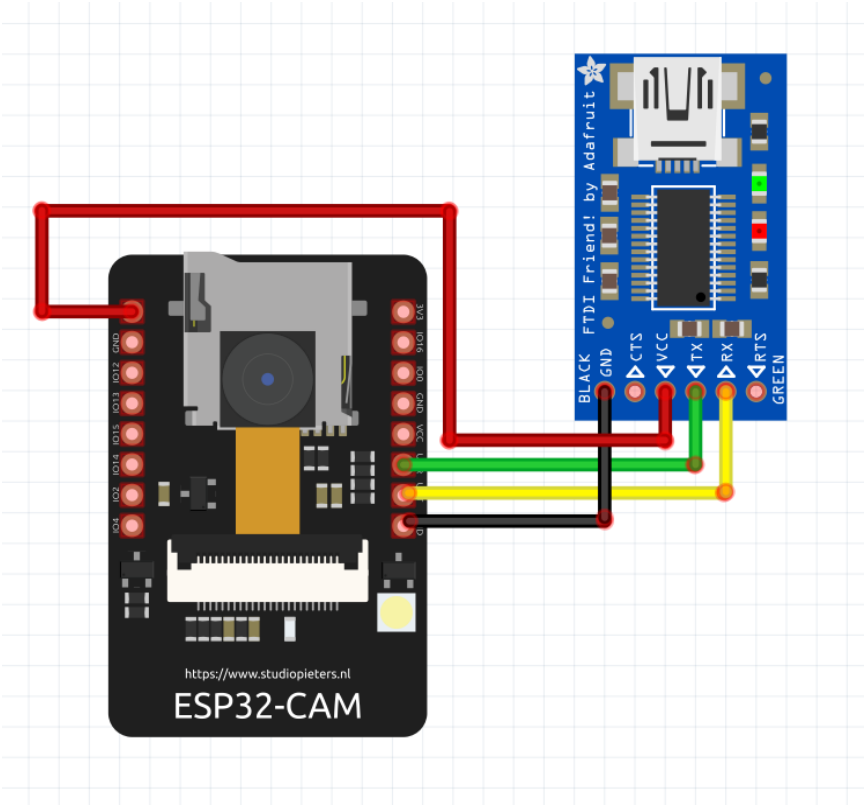
1. Firstly, connect the Camera Sensor with the FPC connector of ESP32-CAM , as shown:

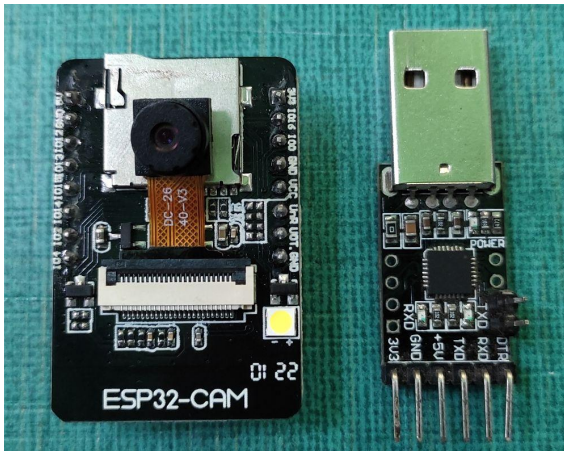


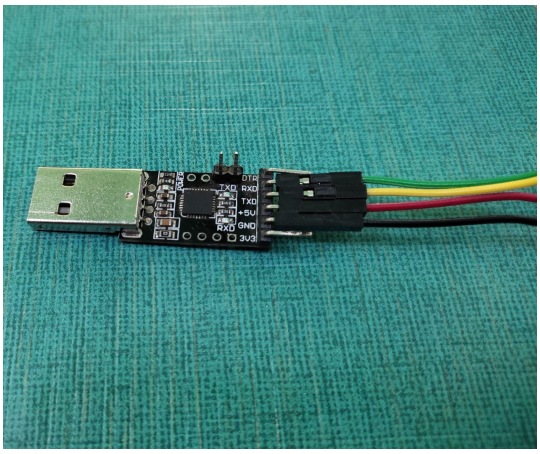


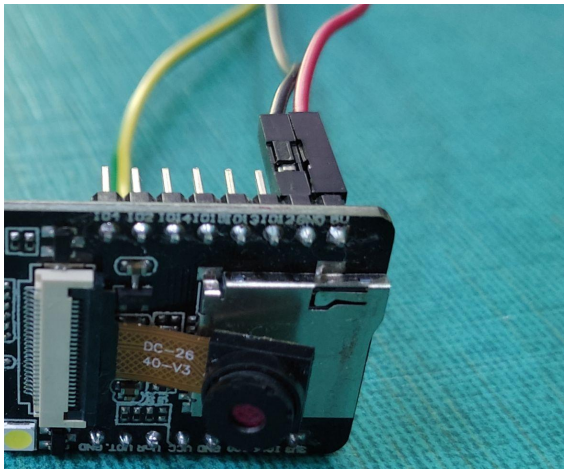


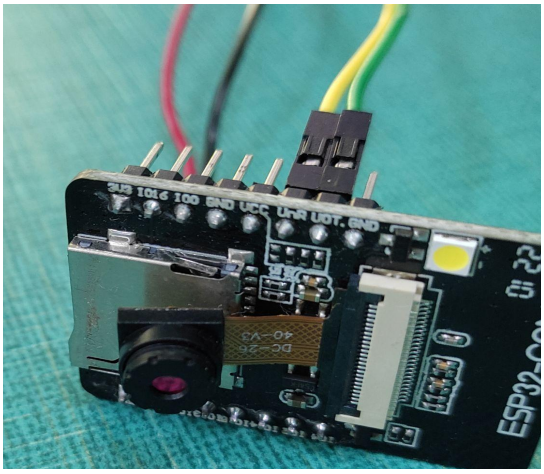
2. Connect ESP32 - CAM with USB-to-TTL converter by using female-to-female jumper wires, as shown:



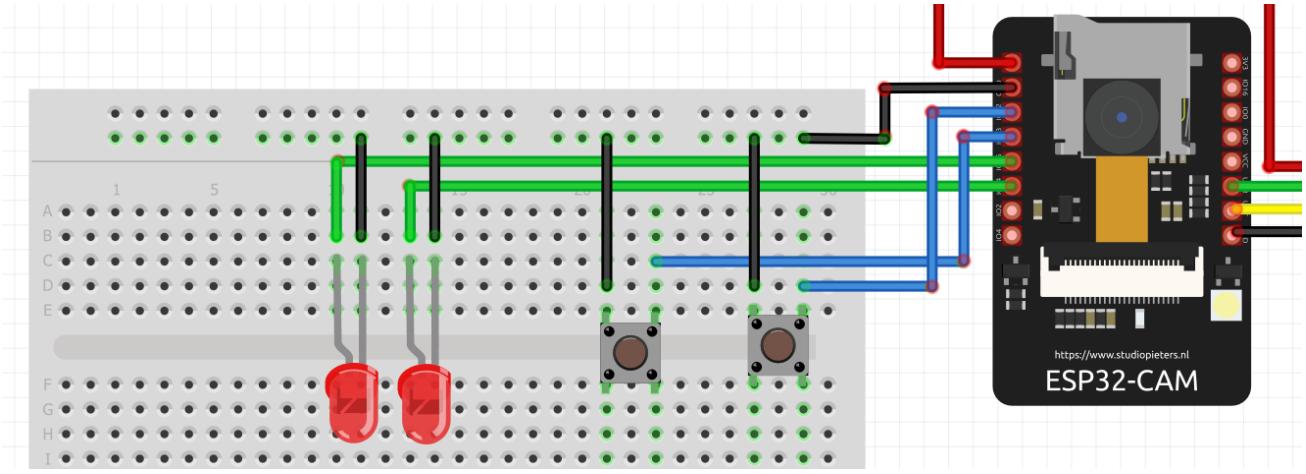








3. Connect Leds and Push Buttons with the above circuit, by using breadboard and jumper-wires.



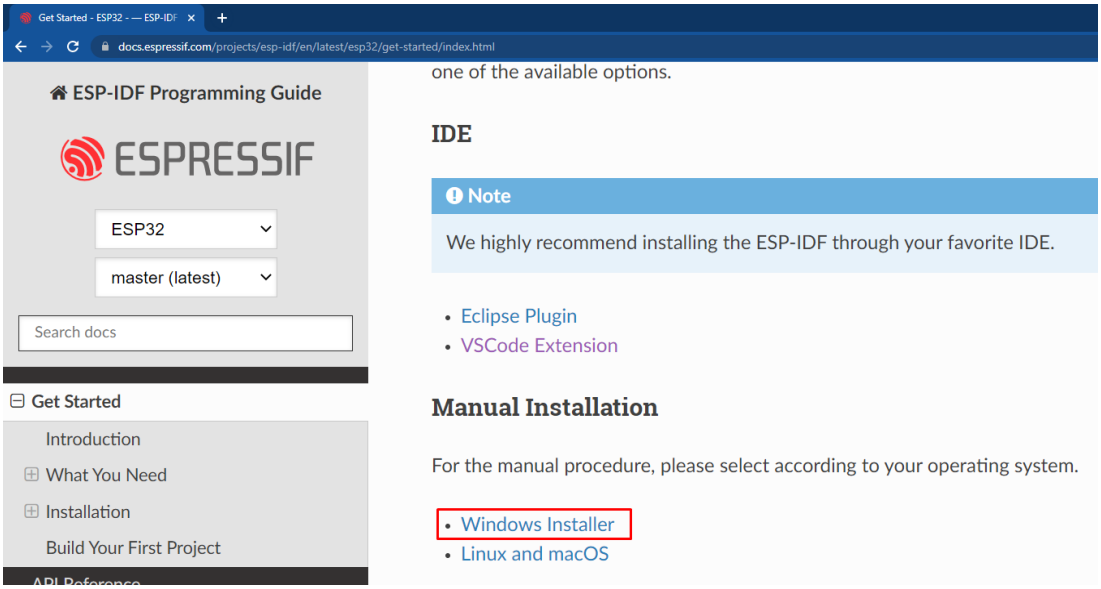
4. Finally, connect the USB-to-TTL converter with your PC.

SOFTWARE SETUP

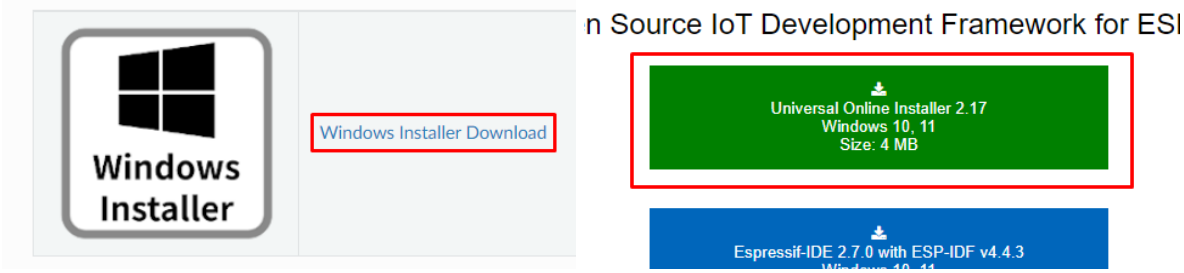
- Install ESP-IDF in Windows
- Installing VS Code
- Integrating ESP-IDF with VS Code
- Setting Up Firebase

1. ESP-IDF Installation in Windows:

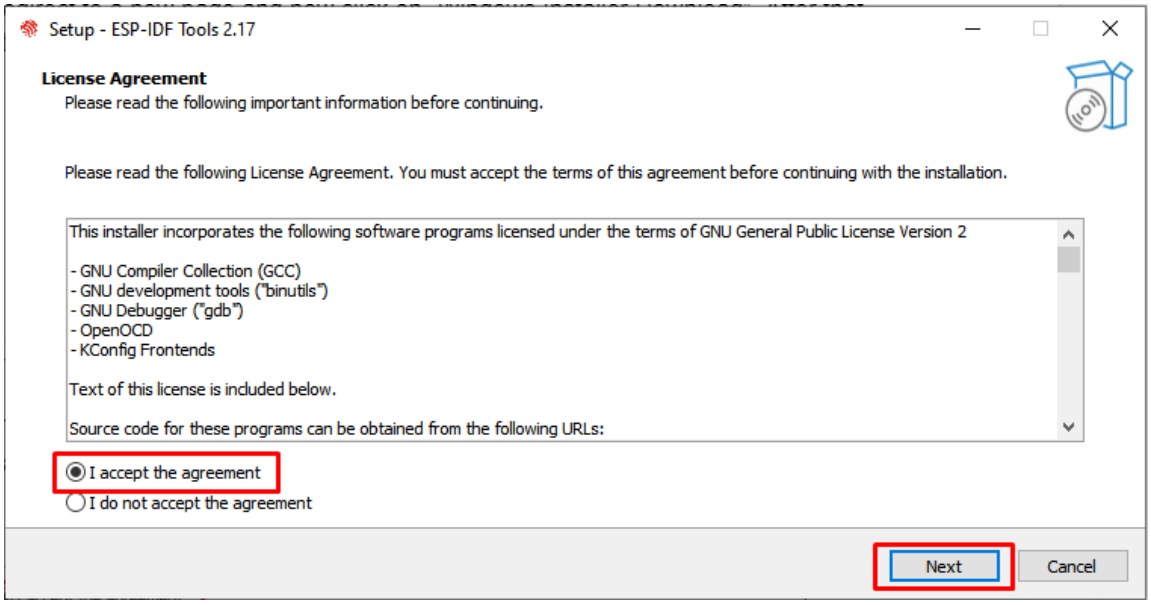
Opens the official ESP32 '[getting started](#)' webpage by Espressif. Scroll down until you reach the Manual Installation section. Here click '**Windows Installer**.'



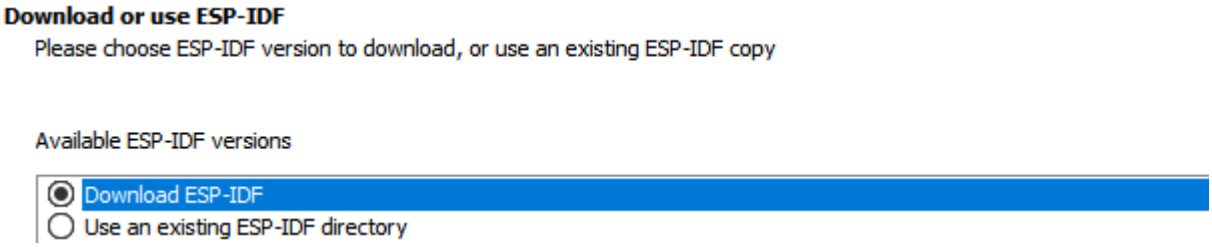
You'll redirect to a new page and now click on "**Windows Installer Download**". After that select the "**Universal Online Installer**".



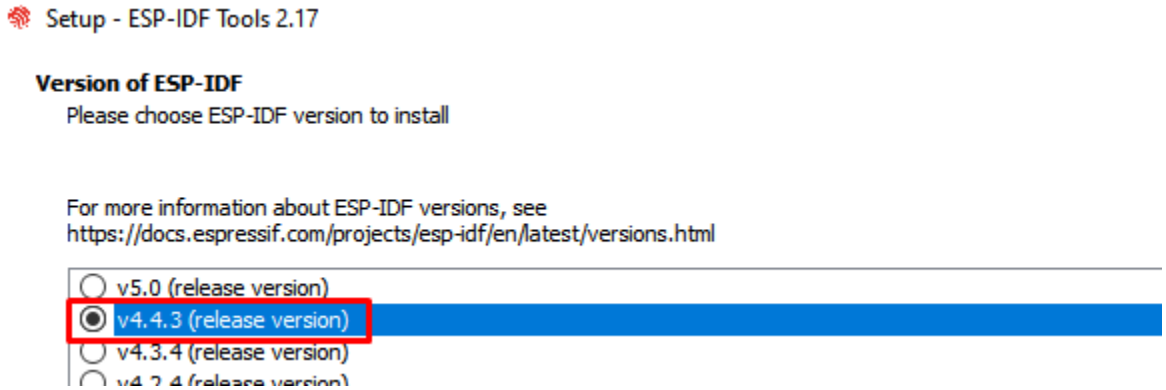
Now, open the downloaded .exe file, and begin esp-idf installation.



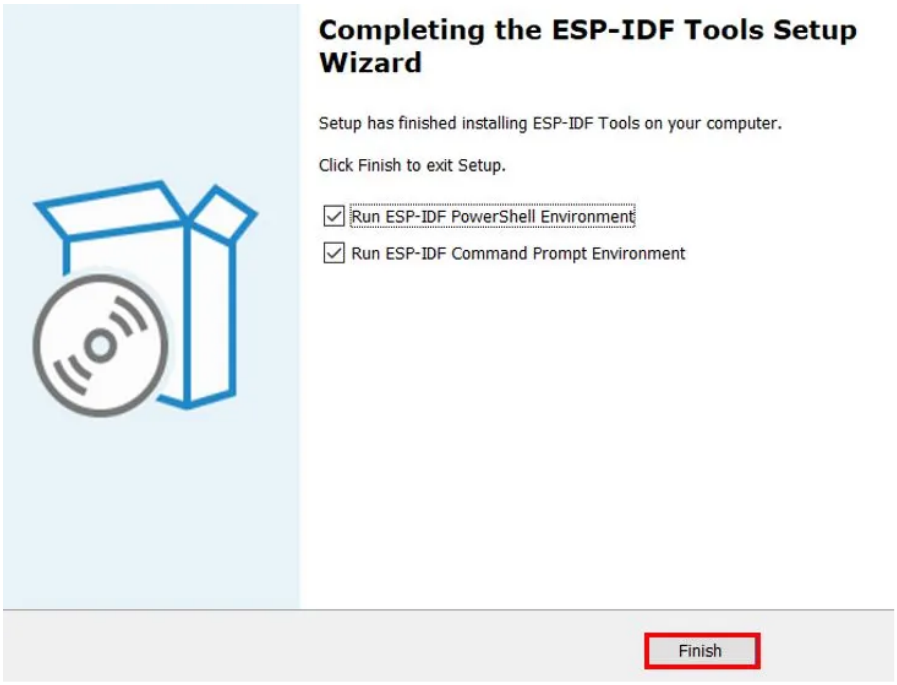
Click '**Next**' on Pre-Installation Check. After that select "**Download ESP-IDF**".



Select the "**V4.4.3**(release version)".

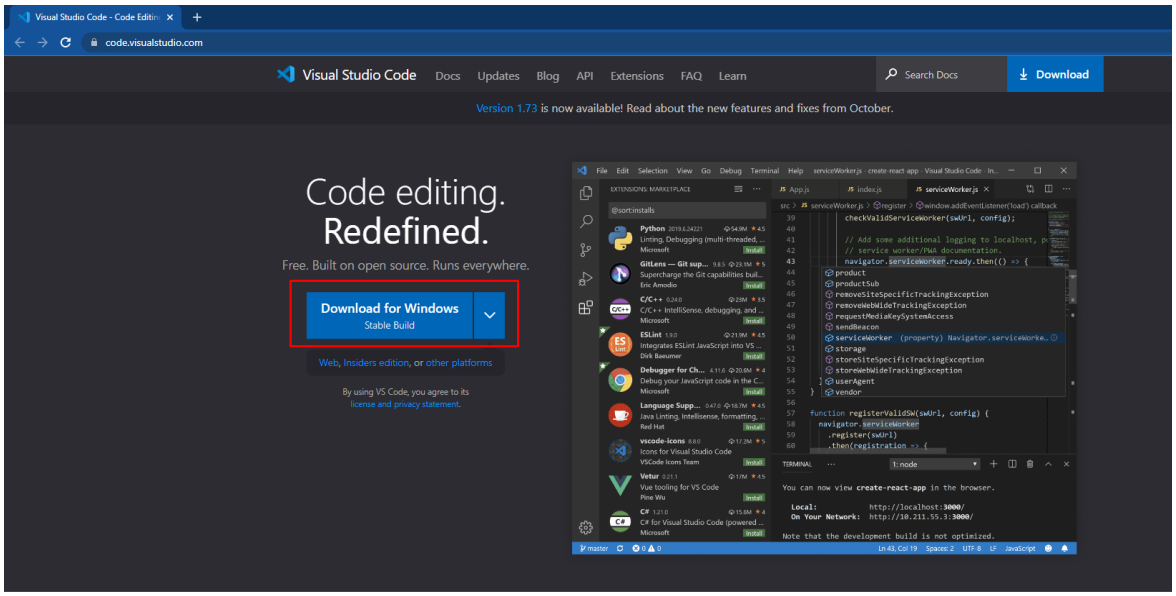


In the "Select Components" tab, remain it as it is and click "Next". Now the installation begins. Click the **Finish button** to exit the setup.

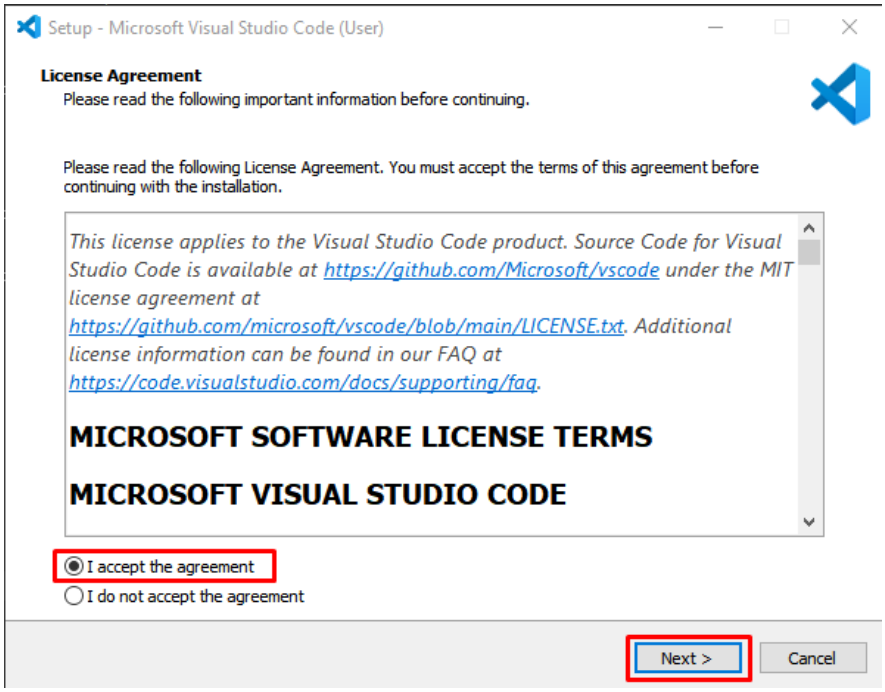


2. Installing VS Code:

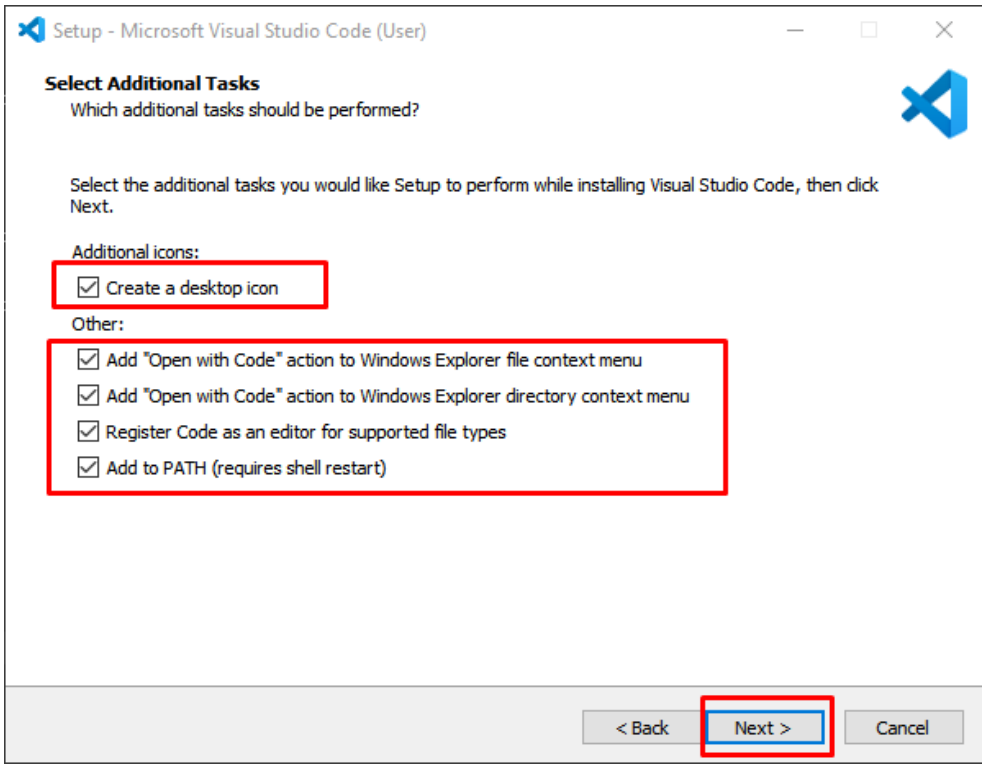
To download and install the visual code editor IDE, visit the official website of vs-code.



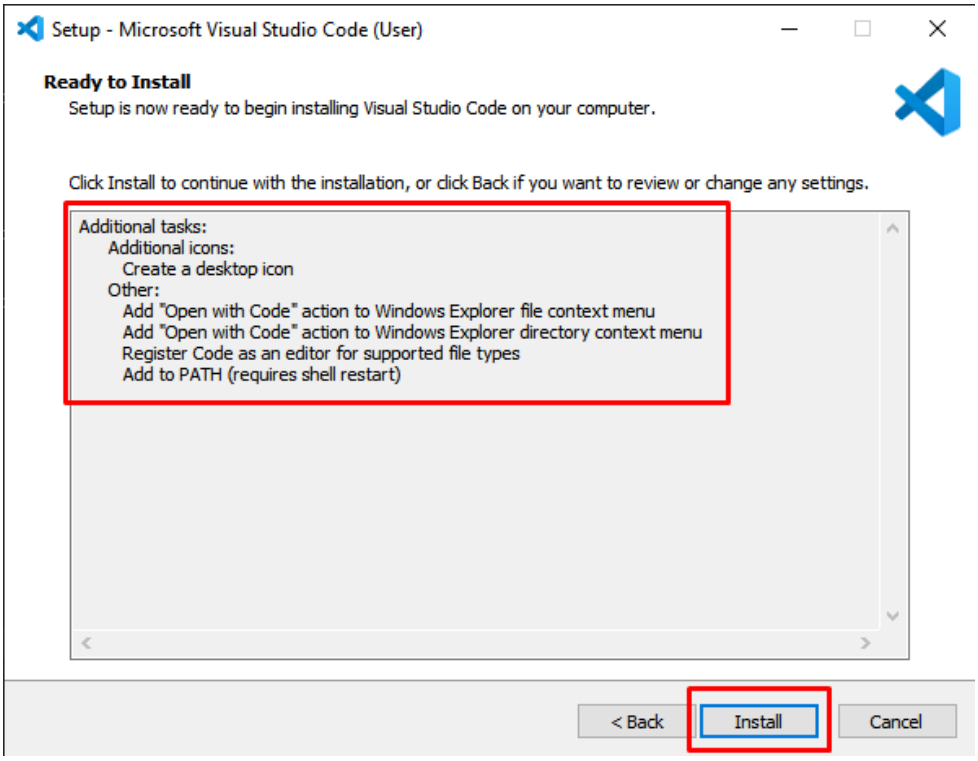
After the download is complete, click it. The following License Agreement window opens up. Here you will accept the agreement and then press **'Next.'**



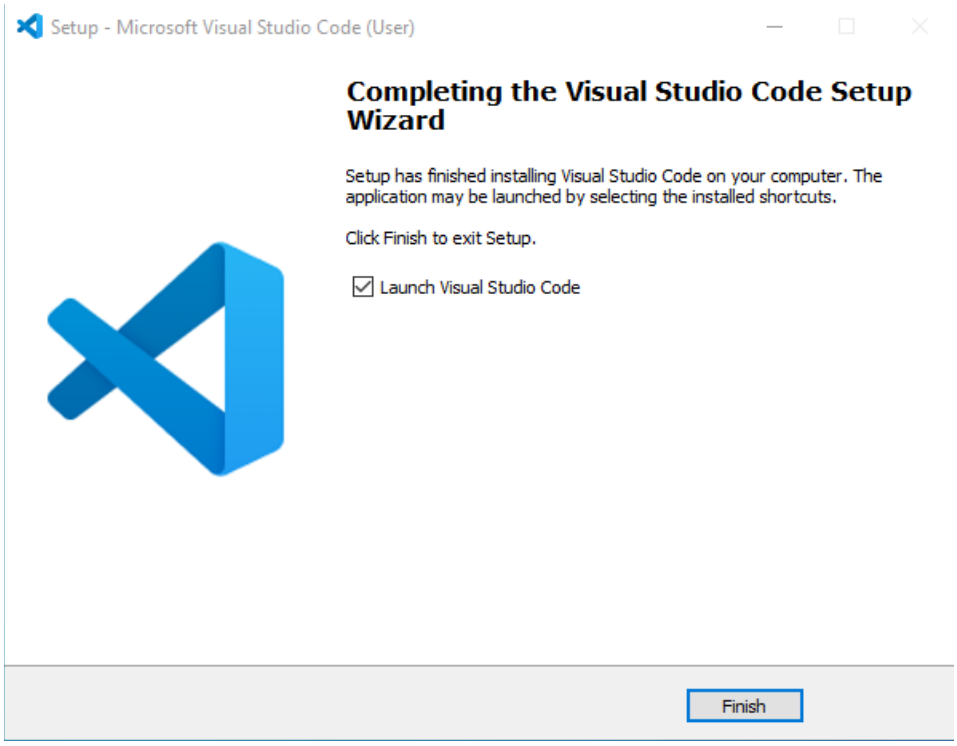
Additionally, you can also select additional tasks according to your needs. After that click '**Next**'.



Now the setup is ready to install. The red block highlights the options we ticked on our way. Finally, click the "**Install**" button.

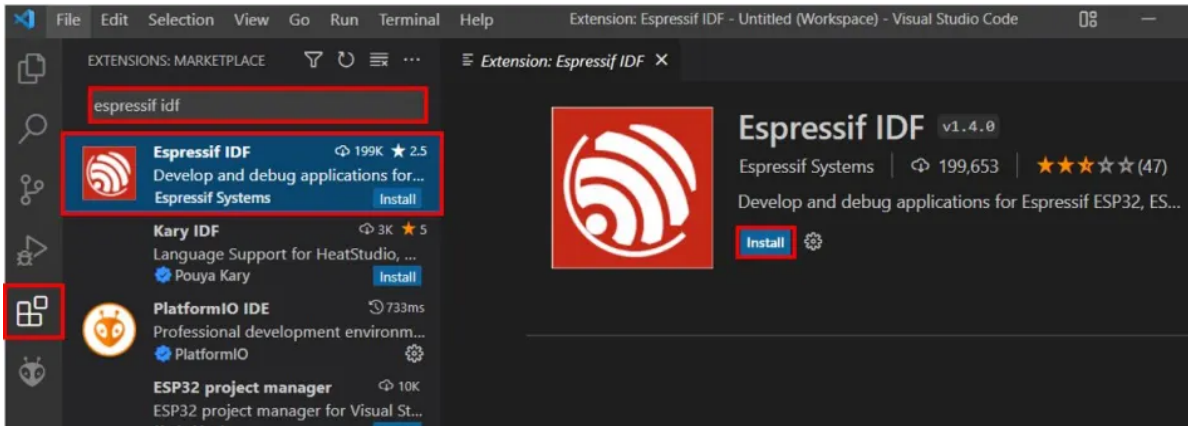


VSCode installation is done.

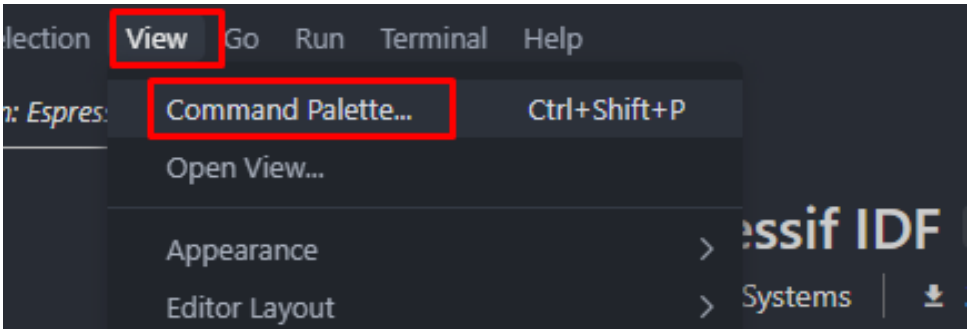


3. Integrating ESP-IDF with VS Code:

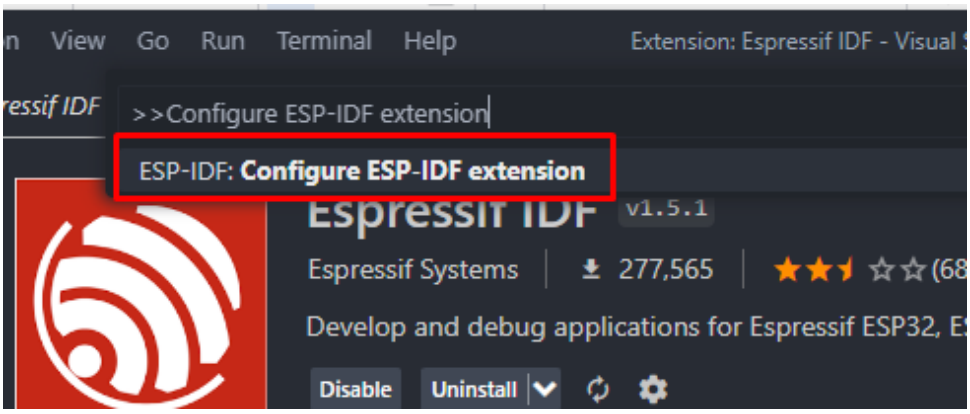
Open Vs-code extensions by pressing Ctrl + Shift + X or click its icon. Type ‘espressif idf’ in the search bar and install it.



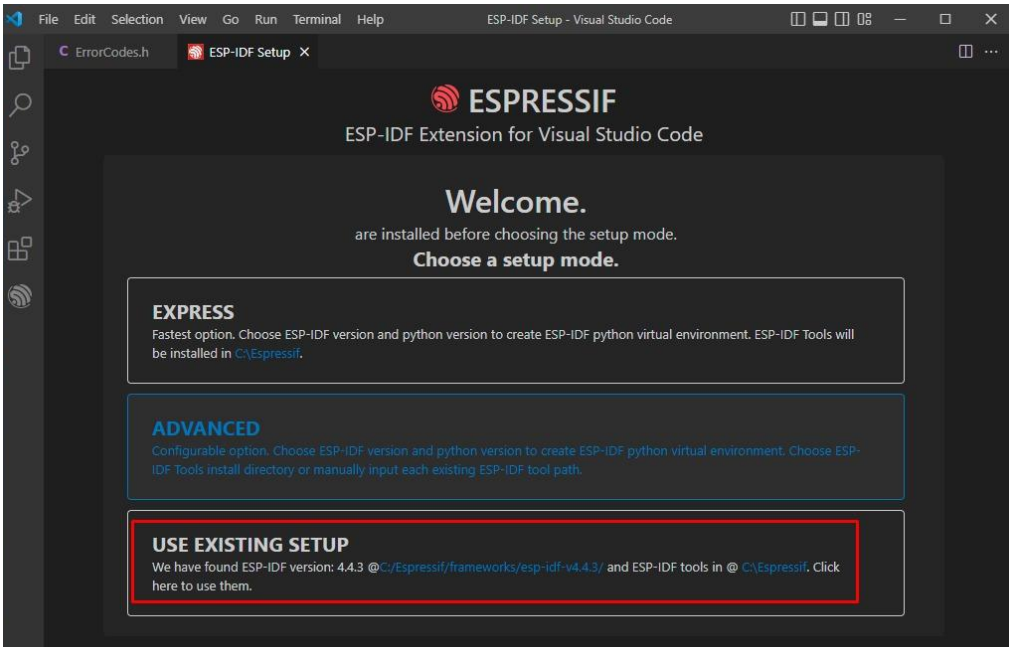
Go to **View->Command Palette**,



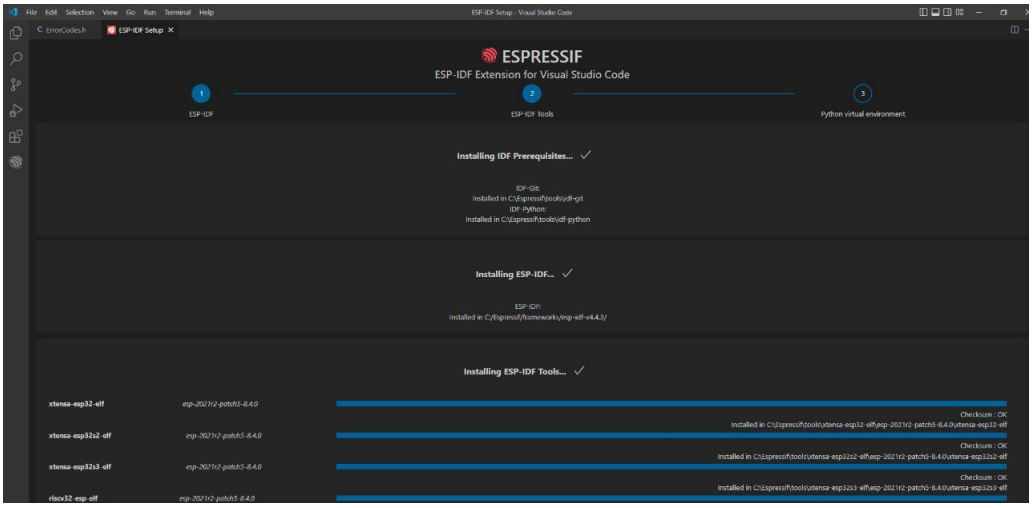
Type “**Configure ESP-IDF extension**” and Select the option.



Select the “**Use Existing Setup**”



VS-code fetched the already installed ESP-IDF.



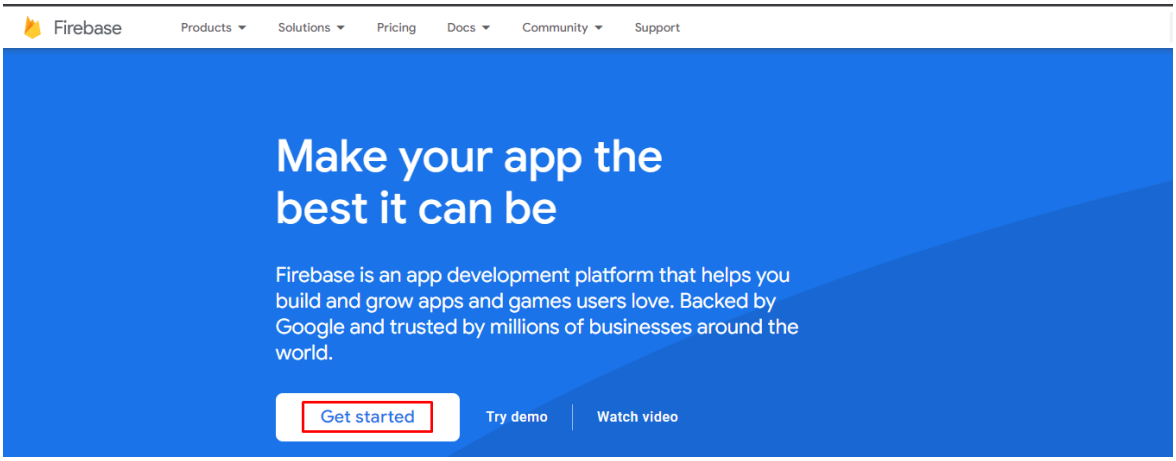
VS-code is configured successfully.



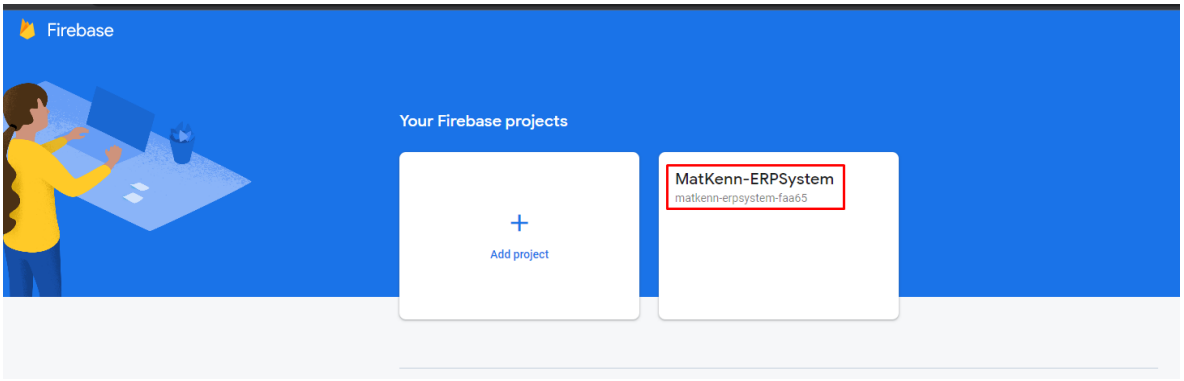
4. FireBase Setup:

Now, setup the firebase by following these steps;

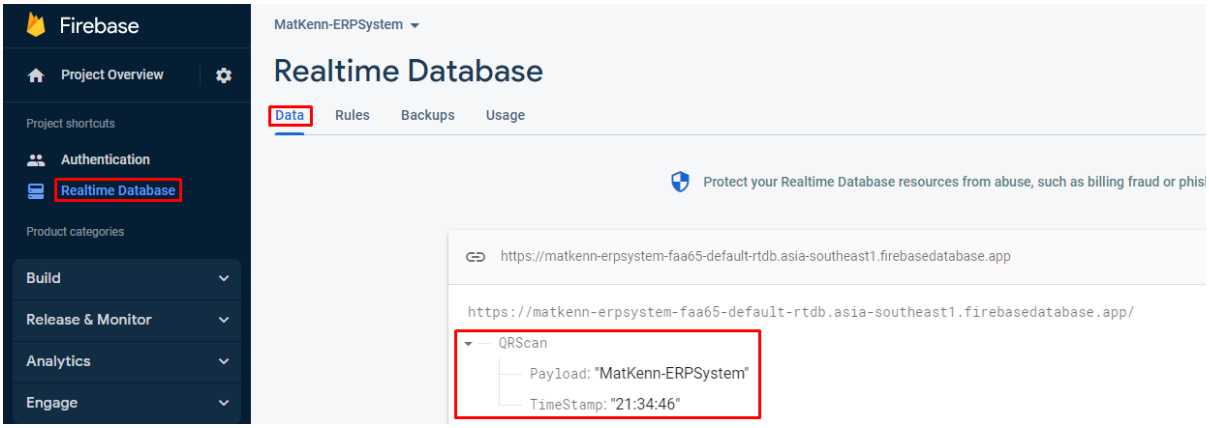
Open the official Firebase website and click on “**Get Started**”.



Sign-in with the provided credentials, and the “**Firebase Projects**” page will open. Now click on “**MatKenn-ERPSystem**”.



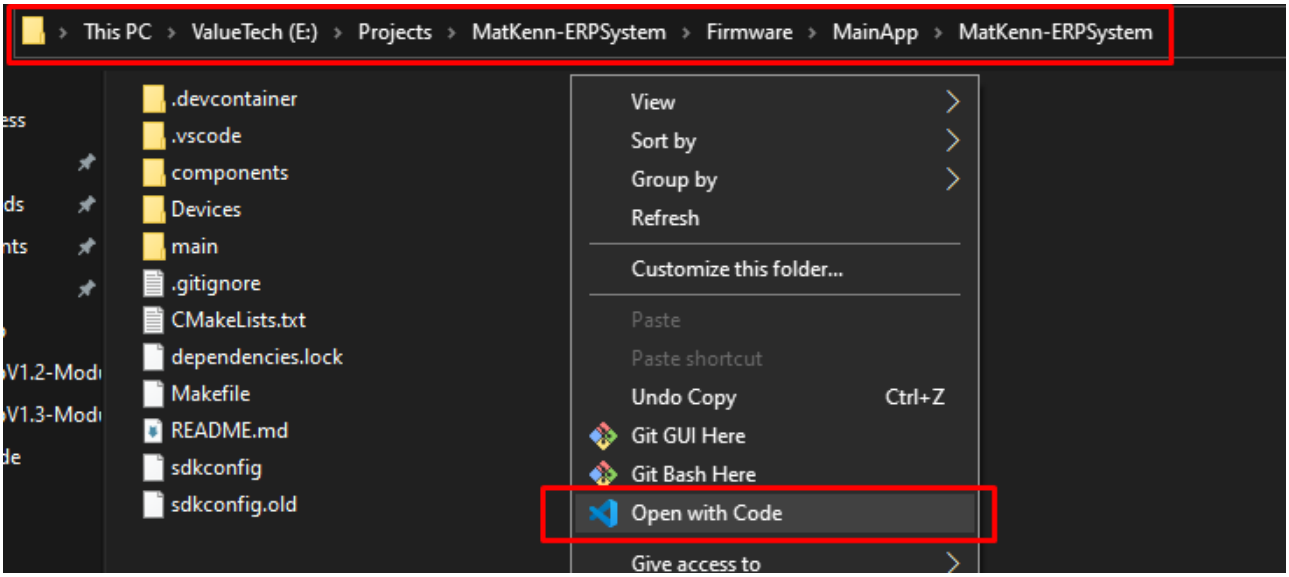
Go to **Realtime Database->Data**, and here you can access the qr-scanner data.



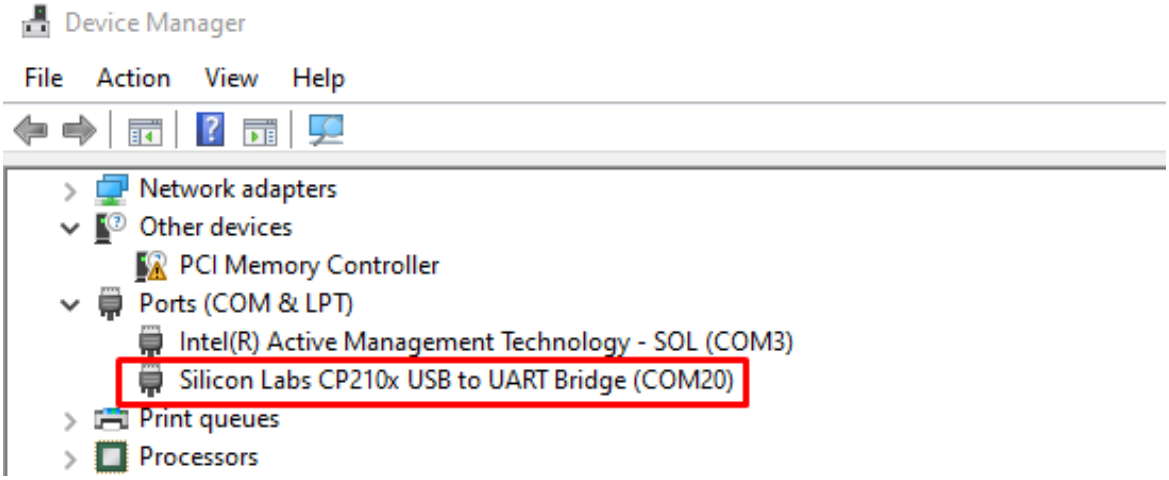
BUILDING, FLASHING & MONITORING (FIRMWARE):

1. Project Build:

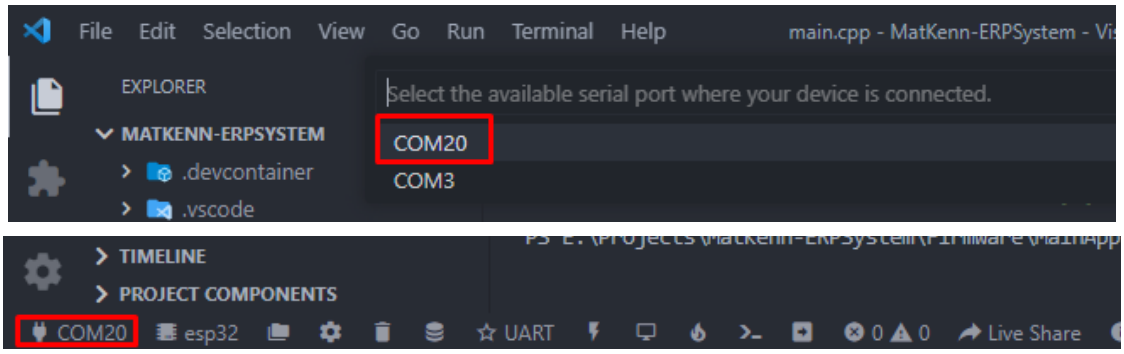
Go to your project directory and open the project in VS code.



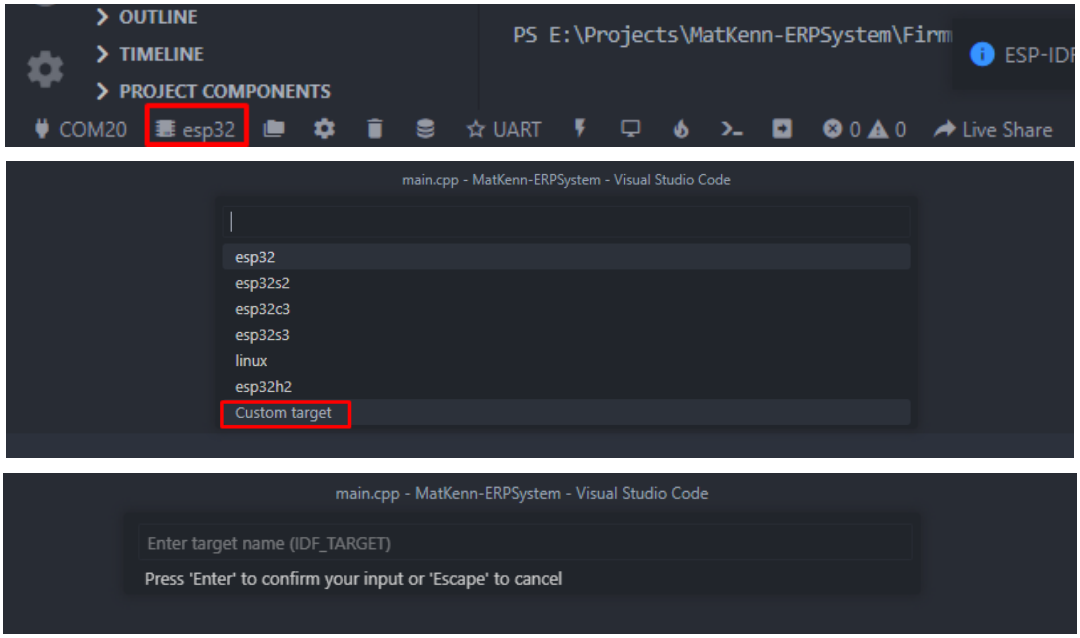
Go to "**Device Manager**" and check your ESP32-Cam **COM Port**. (make sure your board is connected).



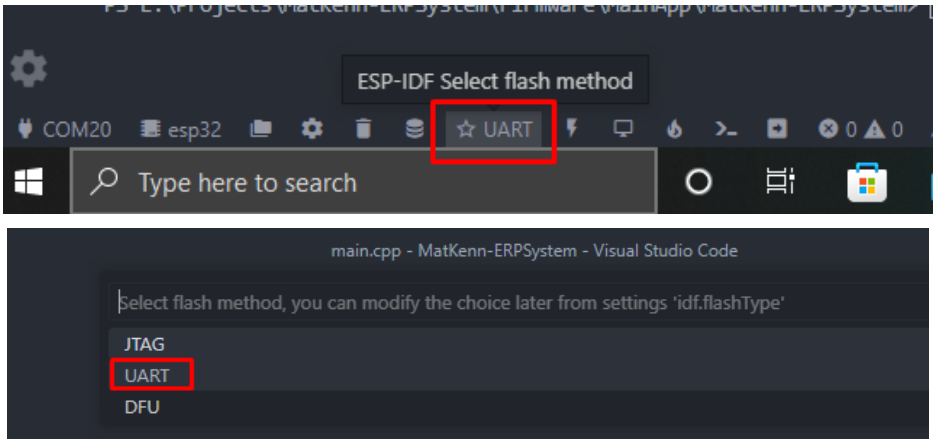
Now, Select this COM Port on Vs Code.



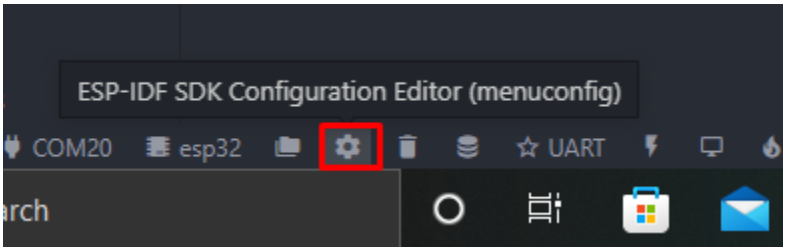
Select "**Custom target**" as your device target and press ESC.

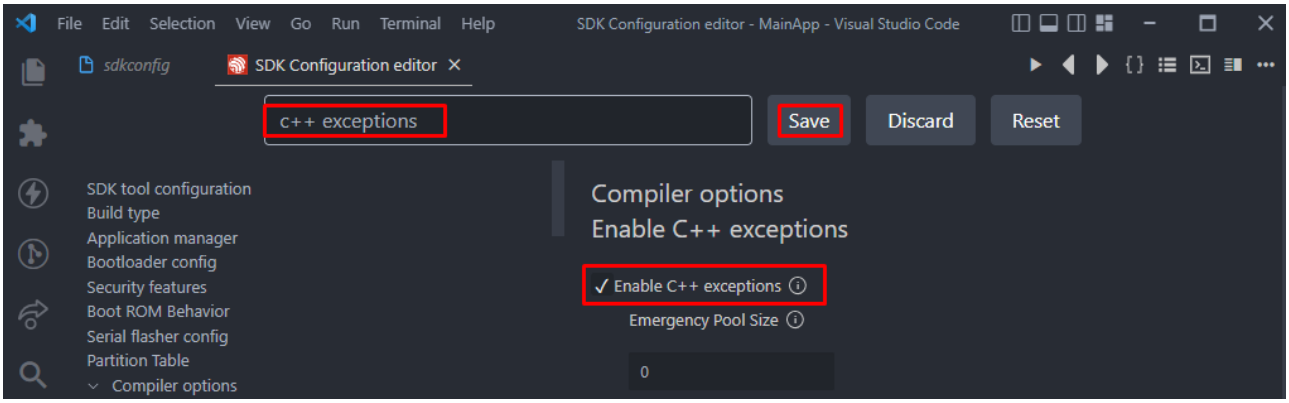


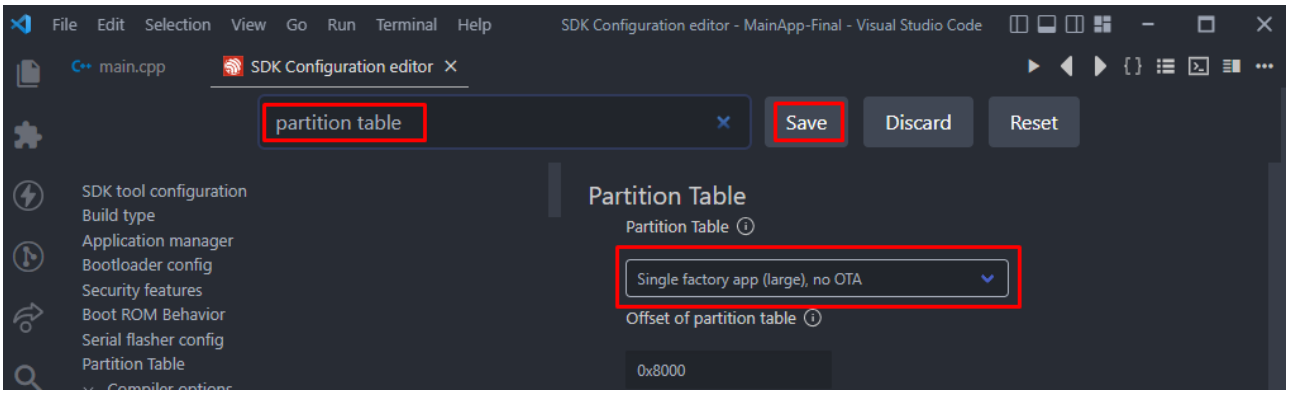
Select "**UART**" in flash method.



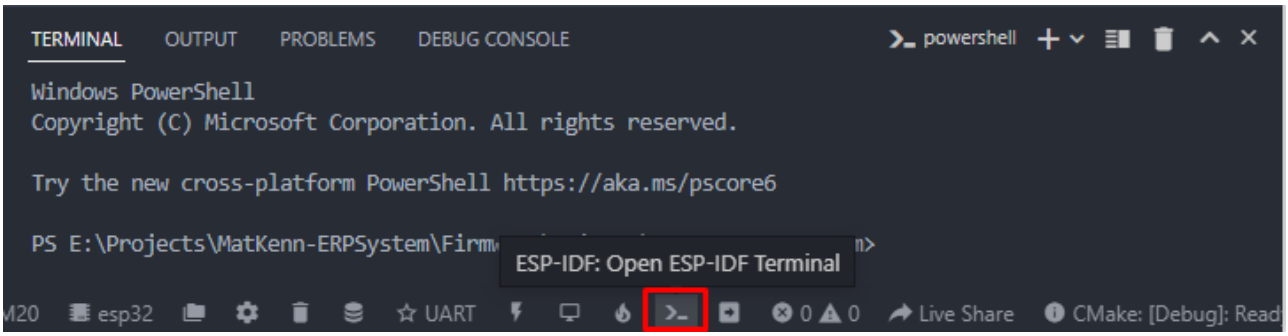
Select the “**menuconfig”** and change the following parameters & click save.

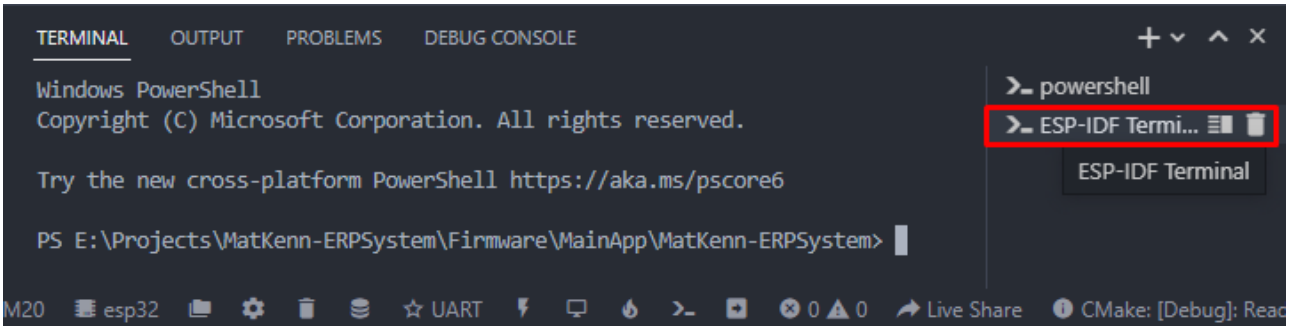




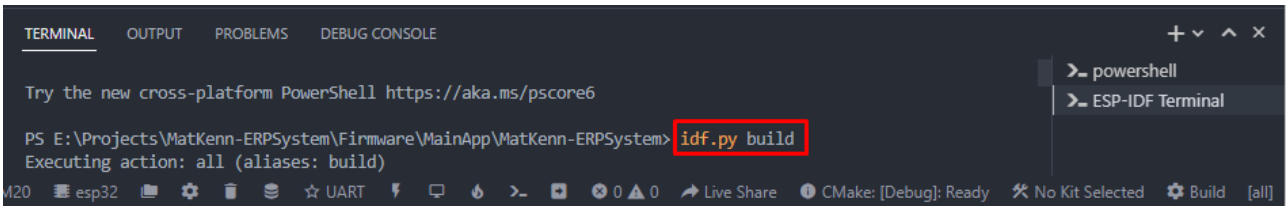


Now open the esp-idf terminal, and **Build the Project**.





Run the following command in ESP-IDF terminal, " **idf.py build** "

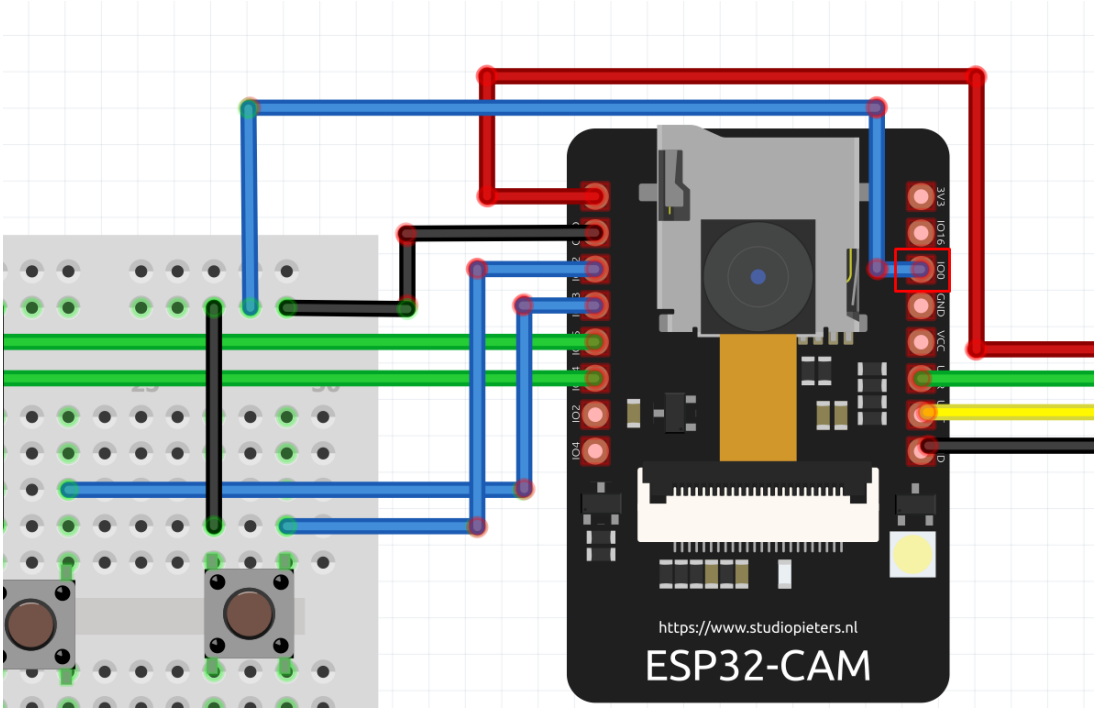


The project build is complete and ready to flash.

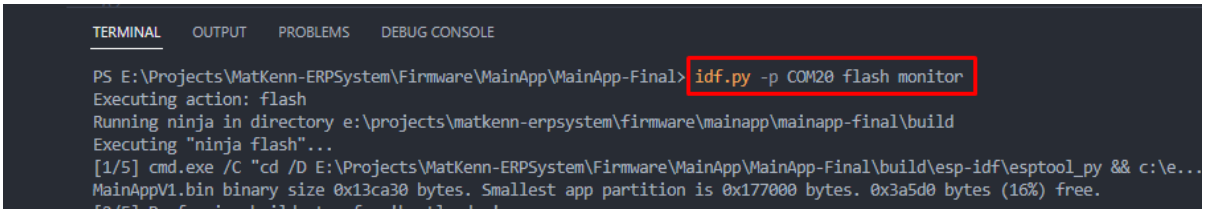


2. Flashing:

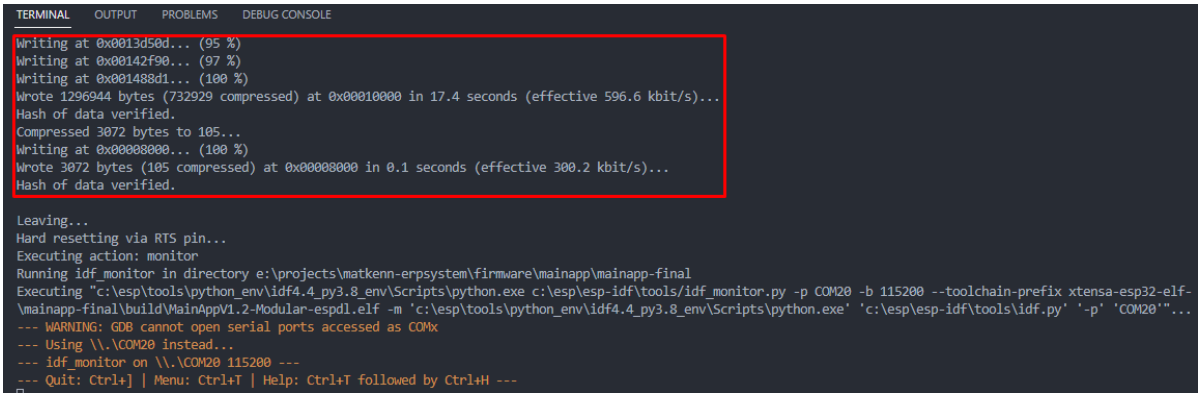
First you have to connect the **GPIO0** to GND and press the reset button of the ESP32-Cam board.



Now, run this command on the esp-idf terminal, “ **idf.py -p COM20 flash monitor**”



Flashing is done. Remove “GPIO0” from “GND” and press the reset Button of the board.



You can monitor the logs on the terminal also;

